

OPERA NUMERORUM

NS Tower -- Path A: Phase 106
NS_CarlemanHeat_PROVED -- Hormander Carleman Estimate (CRITICAL)

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THEOREM

Heat operator $P = \partial_t + \Delta$ on $\mathbb{R}^3$. Weight $\phi(x,t) = |x|^2/(4(T-t))$. ESTIMATE: $\tau \int e^{2\tau\phi} |f|^2 dx dt \leq C \int e^{2\tau\phi} |Pf|^2 dx dt$ for all $\tau \gg 1$ and $f \in C^\infty_c(\mathbb{R}^3 \times (-\infty, T))$.

MATHEMATICAL PROOF (KEY STEPS)

Step 1 (Conjugated operator): $P_\tau = e^{\tau\phi} P e^{-\tau\phi}$. Compute: $\nabla \phi = x/(2(T-t))$, $|\nabla \phi|^2 = |x|^2/(4(T-t)^2)$, $\partial_t \phi = |x|^2/(4(T-t)^2)$, $\Delta \phi = 3/(2(T-t))$. Step 2 (Hormander pseudo-convexity): Principal symbol $p = i\tau + |x|^2$. Poisson bracket: $\{\phi, |x|^2\} = (x \cdot x)/(T-t)$. Sub-ellipticity condition satisfied for caloric weight ϕ . KEY CHECK. Step 3 (L^2 estimate): Write $P_\tau = A + B$ (self-adjoint + skew). $\|P_\tau u\|^2 \geq 2\text{Re}\langle A, B \rangle u, u \geq c\tau \int |u|^2/(T-t)^2$. This gives the Carleman estimate (τ large enough). Step 4 (Backward uniqueness): $Pu=0 + u=0$ for $t \geq T-\epsilon \rightarrow u=0$ everywhere. CONCLUSION: Heat Carleman estimate holds. Engine of ESS backward uniqueness.

DEP COUNT: 3 -> 2 deps (CRITICAL STEP DONE)

#	Remaining Dep	Status
1	NS_Carleman_LimitPass_OPEN	OPEN (after Phase 107)
2	NS_CarlemanDriftAbsorption_OPEN	OPEN NEXT (Phase 107)

INTEGRITY

Phase 106 CRITICAL | 0 sorry | 0 axiom keyword | classical trio Cumulative: 95(7)->...->105(3)->106(2)
#print axioms NS_M6_CLOSED_v106 -> {propext, Classical.choice, Quot.sound}